


# Drug Consumption Risk Modelling Using Bayesian Networks and Influence Diagrams

ECS773 - Bayesian Decision & Risk Analysis

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# 1 Problem Definition & Variable Design

Drug consumption represents a significant public health concern, with the number of people using illicit drugs having risen by over 20% in the past decade (UNODC, 2024). This project will model the causal pathway from demographic and personality factors to drug consumption outcomes, using a Bayesian Network (BN) extended into an Influence Diagram (ID), applied to the UCI Drug Consumption dataset, publicly available on Kaggle.

The selected domain resides in public health risk assessment. The chosen problem being addressed is: *Given an individual's demographic and personality profile, how should a policymaker distribute intervention resources that minimise drug-related harm at the lowest cost?* This involves uncertainty over individual risk and competing costs of intervention methods.

A BN models uncertainty and can tell you the probability that someone becomes a regular cannabis user given their personality profile, but it cannot optimise decisions. IDs extend on BNs by adding decision nodes, which model the decision-making process and utility nodes, that quantify the value of outcomes in terms of costs and benefits, thus leading to the Maximum Expected Utility (MEU). Drug intervention follows this structure, where BNs have been applied successfully to substance use disorder (Weinstein et al., 2009), validating the probabilistic modelling approach in this domain.

This project adopts a hybrid approach, combining domain knowledge for structure, data learned parameters using pyAgrum's *BNLearner* function and two main theoretical frameworks: gateway drug theory, where cannabis use is associated with subsequent harder drug use (National Institute of Justice, 2018) and Big Five personality research (Kotov et al., 2010), which found that substance use disorders are linked to neuroticism, disinhibition, and low conscientiousness. More recently, high sensation-seeking has been frequently associated with initiation of recreational substance abuse (Maharjan et al., 2025), driven by desire for stimulation and positive affect.

The BN nodes are comprised of:

- **Age, Gender, Education** – demographic root nodes
- **Neuroticism** – emotional instability linked to coping-substance abuse
- **Openness** – novelty-seeking trait
- **SensationSeeking** – direct predictor of substance use
- **Cannabis** – central gateway node
- **Cocaine, LSD** – outcome nodes
- **Intervention, EducationCampaign** – decision nodes
- **HealthOutcome, CostOfIntervention** – utility nodes

## 2 Network Structure and Justification

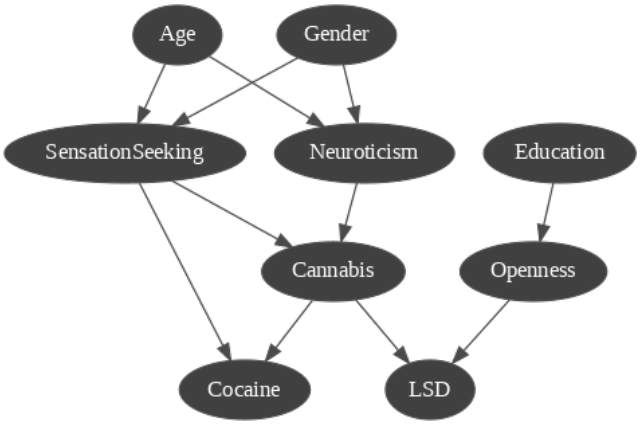



Figure 1: BN Structure

Figure 1 presents the BN structure generated by pyAgrum. The network consists of nine chance nodes in a three-layer architecture: demographic root nodes (Age, Gender, Education), personality mediators (Neuroticism, Openness, SensationSeeking), and drug outcome nodes (Cannabis, Cocaine, LSD). Nine variables were selected from the full dataset to balance model interpretability whilst maintaining coverage of key demographics and outcomes relevant to the decision problem.

The network structure was defined manually in pyAgrum using *addArc*, encoding motivated causal relationships described in Question 1. Automated structure learning algorithms such as Hill Climbing (HC) were considered but rejected, as they optimise for statistical fit rather than causal plausibility, risking misleading arc directions. Prior to parameter learning, variables were discretised - personality scores reduced to Low/Medium/High, drug use collapsed from seven classes to three, and age/education simplified into broader categories to maximise clear interpretability. Parameters were then estimated from 1,884 observations via the MLE using *learnParameters*.

# 3 Parametrisation of Conditional Probability Tables (CPTs)

| Age | | |
|---|---|---|
| Young | Middle | Old |
| 0.5966 | 0.3445 | 0.0589 |

| | | SensationSeeking | | |
|---|---|---|---|---|
| Gender | Age | Low | Medium | High |
| F | Young | 0.3804 | 0.3559 | 0.2637 |
| | Middle | 0.5822 | 0.3092 | 0.1086 |
| | Old | 0.5294 | 0.3333 | 0.1373 |
| M | Young | 0.1551 | 0.3777 | 0.4671 |
| | Middle | 0.3586 | 0.4379 | 0.2034 |
| | Old | 0.5833 | 0.3833 | 0.0333 |

| | | Cannabis | | |
|---|---|---|---|---|
| Neuroticism | SensationSeeking | NonUser | Occasional | Regular |
| Low | Low | 0.6603 | 0.1579 | 0.1818 |
| | Medium | 0.2500 | 0.2634 | 0.4866 |
| | High | 0.1339 | 0.1417 | 0.7244 |
| Medium | Low | 0.6049 | 0.2413 | 0.1538 |
| | Medium | 0.3211 | 0.3049 | 0.3740 |
| | High | 0.1340 | 0.2320 | 0.6340 |
| High | Low | 0.4253 | 0.3218 | 0.2529 |
| | Medium | 0.1900 | 0.3122 | 0.4977 |
| | High | 0.0690 | 0.2611 | 0.6700 |

| | | LSD | | |
|---|---|---|---|---|
| Cannabis | Openness | NonUser | Occasional | Regular |
| NonUser | Low | 0.9934 | 0.0066 | 0.0000 |
| | Medium | 0.9706 | 0.0210 | 0.0084 |
| | High | 0.9474 | 0.0526 | 0.0000 |
| Occasional | Low | 0.8497 | 0.1373 | 0.0131 |
| | Medium | 0.7296 | 0.2398 | 0.0306 |
| | High | 0.6328 | 0.2734 | 0.0938 |
| Regular | Low | 0.6875 | 0.2500 | 0.0625 |
| | Medium | 0.5145 | 0.3441 | 0.1415 |
| | High | 0.3562 | 0.3890 | 0.2548 |

Figure 2: CPTs for Age (root), SensationSeeking (2 parents), Cannabis (intermediate), LSD (leaf)

Figure 2 presents the CPTs for four selected nodes. As Question 2 mentioned, no CPTs were manually constructed, and these figures are derived from the learnt parameters. **Age** is a root node with no parents. Its CPT is a simple prior probability distribution across three states (Young, Middle, Old), reflecting the marginal distribution of age in the dataset. **SensationSeeking** has two parents (Age, Gender), requiring 18 parameters. Its CPT captures how sensation seeking varies across age/gender combinations, reflecting demographic influences on personality. **Cannabis** is an intermediate node with 3 parents (Neuroticism, SensationSeeking, Openness), requiring 81 parameters, the most complex CPT in the network. It represents the conditional probability of cannabis usage given personality trait combinations. **LSD** is a leaf node with two parents (Cannabis, Openness), requiring 27 parameters. Its CPT captures escalation from cannabis use moderated by openness to experience.

The contrast in complexity is significant - from 3 parameters for a root node to 81 for Cannabis illustrates how exponentially large CPTs can grow with the number of parents, motivating the discretisation choices made during preprocessing.

# 4 Observational, Interventional, and Counterfactual Reasoning

## 4.1 Observational Reasoning

| Cannabis | | |
|---|---|---|
| NonUser | Occasional | Regular |
| 0.3271 | 0.2538 | 0.4191 |

Prior $P$(Cannabis)

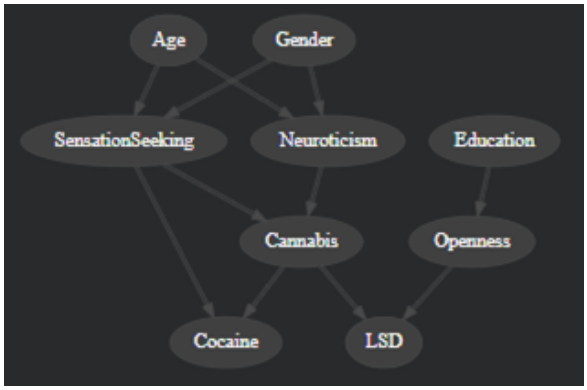
| Cannabis | | |
|---|---|---|
| NonUser | Occasional | Regular |
| 0.1118 | 0.2136 | 0.6746 |

Posterior $P$(Cannabis | Young, Male, High SS)

Figure 3: Observational Reasoning: Prior and Posterior distributions of Cannabis

Figure 3 presents the prior and posterior distribution of Cannabis. Without Evidence $E$, the prior assigns 32.7% NonUser, 25.4% Occasional, and 41.9% Regular. Conditioning on Age=Young, Gender=Male, SensationSeeking=High shifts the posterior 11.2% NonUser, 21.4% Occasional, and 67.5% Regular. This increase reflects how the combined influence of demographic and personality risk factors propagates through the network.

## 4.2 Interventional Reasoning




Causal Model

| Cannabis | | |
|---|---|---|
| NonUser | Occasional | Regular |
| 0.1133 | 0.2145 | 0.6722 |

Impact

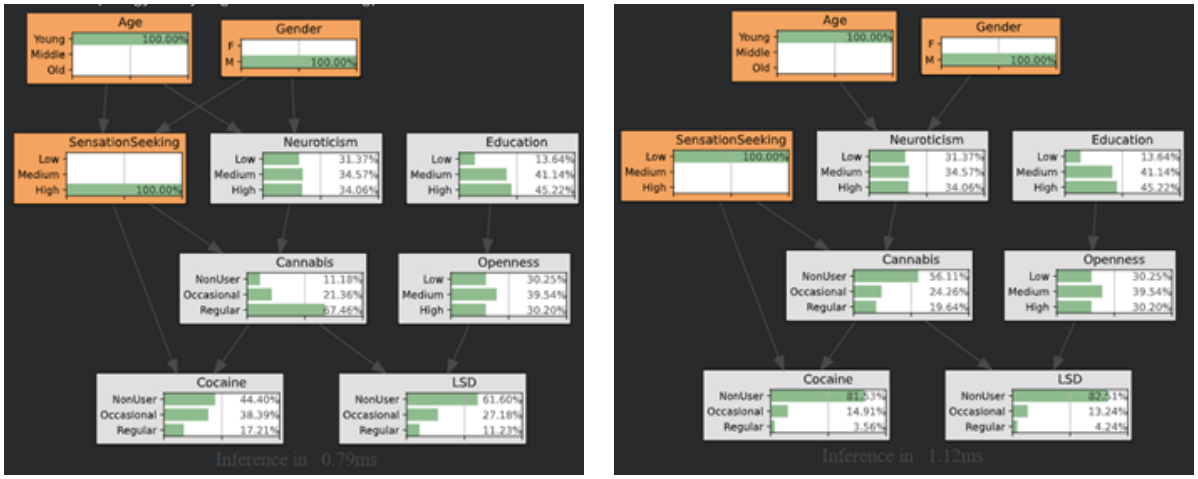
Figure 4: Interventional Reasoning: Causal impact of $do$(SensationSeeking = High) on Cannabis

Figure 4 shows the causal impact of *do(SensationSeeking = High)* on Cannabis. Conditioning on this observation updates belief through the network whilst allowing Age and Gender to continue influencing SensationSeeking. The do-operator however severs all incoming arcs to SensationSeeking and isolates its direct causal relationship with Cannabis. PyAgrum indicated Neuroticism as a backdoor variable (see codebase) and applied it to the following formula:

$$P(\text{Cannabis} \mid do(\text{SensationSeeking})) = \sum_{\text{Neuroticism}} P(\text{Cannabis} \mid \text{Neuroticism}, \text{SensationSeeking}) \cdot P(\text{Neuroticism})$$

The result, P(Cannabis=Regular) = 0.6722, closely matches the observational posterior, suggesting SensationSeeking exerts a strong causal effect on Cannabis use, independent of demographics. Unlike conditioning, which propagates through all active paths including Age and Gender, the do-operator severs these incoming arcs, isolating the direct causal effect of SensationSeeking independent of its demographic causes.

### 4.3 Counterfactual Reasoning




Real World Counterfactual World

Figure 5: Counterfactual Reasoning: Cannabis distribution under real world vs. Low SensationSeeking counterfactual

Figure 5 exhibits a counterfactual scenario: given a Young Male with High SensationSeeking in the real world, what would their cannabis distribution look like had SensationSeeking been Low instead? In this contrast, SensationSeeking was emancipated by removing its incoming arcs from Age and Gender, then fixed to Low. Though most notably, Cannabis=Regular probability decreased by 48%, indicating SensationSeeking as the dominant risk factor in the model, with downstream effects also visible in Cocaine=Regular dropping from 17.2% to 3.6%.

## 5 Bayesian to Influence Diagrams




Figure 6: Influence Diagram extending the BN with decision and utility nodes

Figure 6 extends the BN into an ID by introducing two decision nodes and two utility nodes, enabling optimal policy computation via MEU. **Decision Nodes**

- **Intervention** – represents the decision to offer a targeted drug intervention program to an individual, informed by observed Cannabis and Cocaine risk.
- **EducationCampaign** – infers the decision to deploy a drug-awareness campaign, informed by Age and SensationSeeking that'll be decided after Intervention to emulate natural policy sequencing.

**Utility Nodes**

- **HealthOutcome** – captures health benefits of the intervention decision conditional on Cannabis use. Defined on a scale of +10 (optimal health, no intervention needed) to -6 (regular use, no intervention), reflecting the unnecessary cost of intervening given a NonUser.
- **CostOfIntervention** – highlights the financial burden of each decision combination, from 0 (no action) to -10 (both methods of interventions deployed simultaneously).

# 6 Decision Analysis

## 6.1 Scenario Testing

| **Scenario** | **MEU (mean)** | **Variance** |
|---|---|---|
| Scenario 1: Intervention=Yes, EducationCampaign=No | -3.02 | 6.97 |
| Scenario 2: Intervention=No, EducationCampaign=Yes | -1.48 | 47.78 |
| Optimal Policy: Intervention=No, EducationCampaign=No | 1.52 | 47.78 |

Table 1: MEU results for scenario testing and optimal policy

Table 1 presents the MEU results for two manually fixed decision scenarios alongside the optimal policy. Scenario 1 deploys intervention alone, yielding MEU = -3.02, which indicates intervention costs outweigh health benefits at the population level. Scenario 2, deploying EducationCampaign alone produced a MEU of -1.48, a negative utility reflecting that even the lower-cost campaign does not overcome the cost penalties at the population level.

### 6.2 Optimal Policy

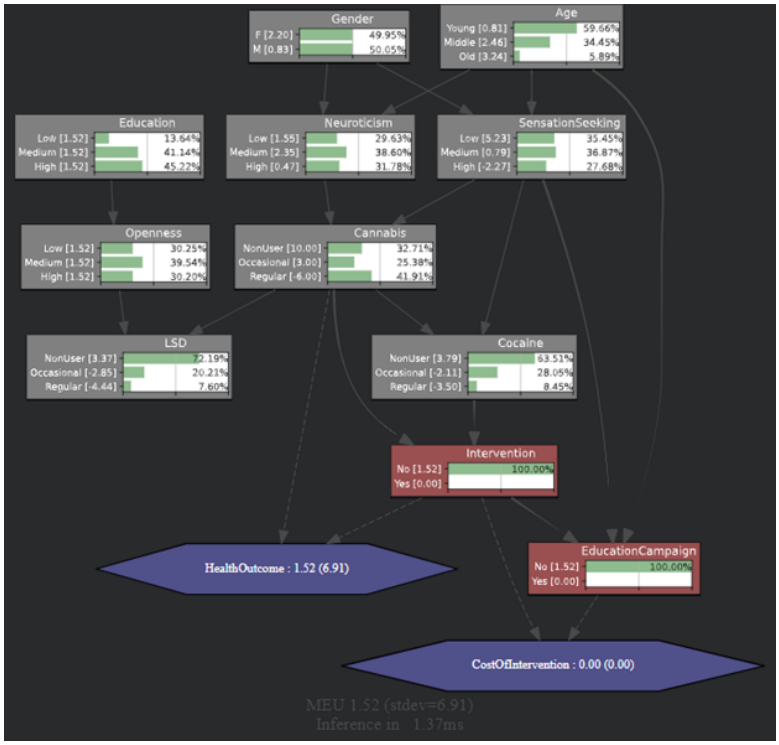



Figure 7: LIMID inference: optimal policy and MEU

Figure 7 presents the optimal policy computed by *ShaferShenoyLIMIDInference*. The MEU of 1.52 is achieved by setting Intervention=No and EducationCampaign=No. This implies that, at the population level, the expected health gains from intervention alone do not outweigh the associated financial costs, given the prior distribution of Cannabis use in the dataset. However, this does not suggest intervention is ineffective, but rather that targeted intervention conditioned on higher-risk profiles would be more appropriate than population-wide deployment, consistent with evidence-based drug policy literature (UNODC, 2024). As demonstrated in Question 4, conditioning on a Young Male with High SensationSeeking yields P(Cannabis=Regular) = 0.675, a profile for which Intervention would become the expected decision. The BN component serves this purpose precisely by identifying which individuals warrant intervention before any decision is made.

## 7 Critical Evaluation

**Strength:** The hybrid construction is the model's primary strength. Manually encoding causal structures from gateway drug theory and Big Five personality research ensures arcs are theoretically motivated, avoiding spurious correlations that pure-data driven structured learning such as HC may recover. Data-learned parameters via MLE then provided empirically grounded CPTs from 1,884 observations, preferred over manual selection across the 9 variables. Ultimately, this combination produced a model that is both causally interpretable and data-informed.

**Weakness:** Utility values assigned to HealthOutcome and CostOfIntervention are not defined on a predetermined scale and have no empirical grounding. No real-world figures (such as NHS program costs) are utilised meaning the MEU result may not accurately reflect genuine policy trade-offs, limiting the models applicability as a real-world decision support tool.

**Improvement:** Therefore, replacing utility values with data-driven estimates would improve the model's validity, grounding the MEU computation in real-world finance and health outcome data.

## References


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